

EE115B *Digital Circuits*

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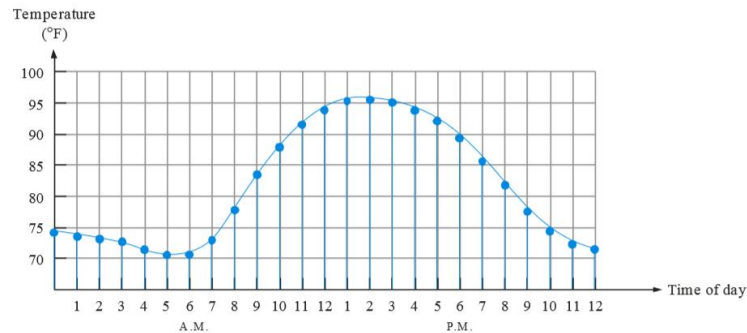
上海科技大学
ShanghaiTech University

Part 2

Introductory Concepts

Analog & Digital Quantities & Advantages?

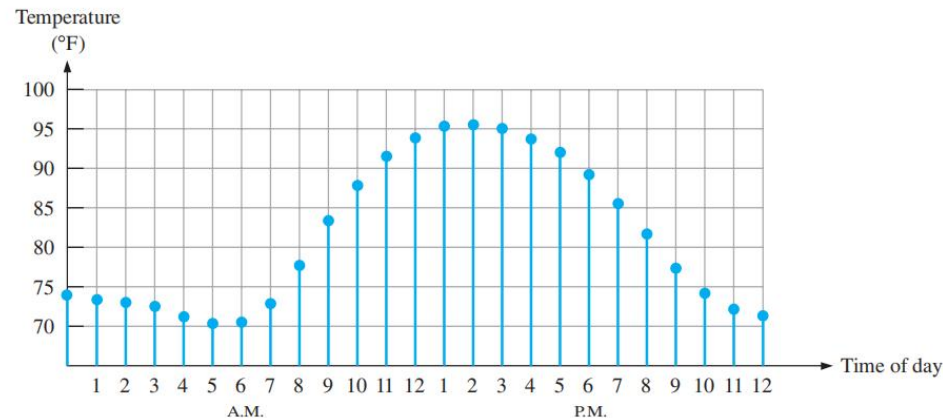
Most natural quantities that we see are **analog** and vary continuously. Analog systems can generally handle higher power than digital systems.



Challenges in analog system:

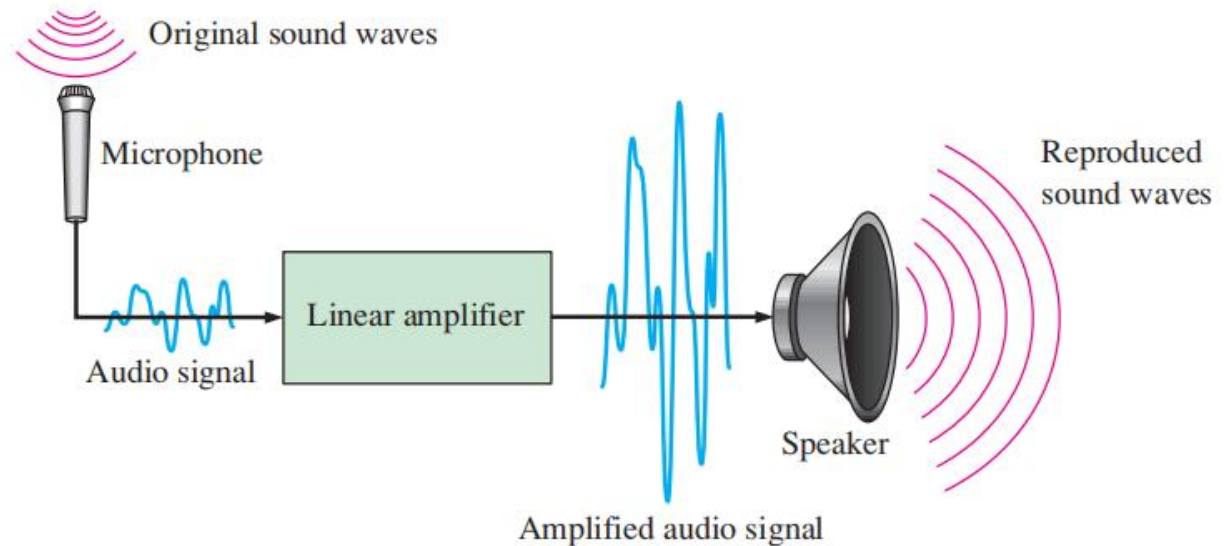
- Storage
- Noise and accuracy
- Computation

Digital systems can process, store, and transmit data more efficiently but can only assign discrete values to each point.

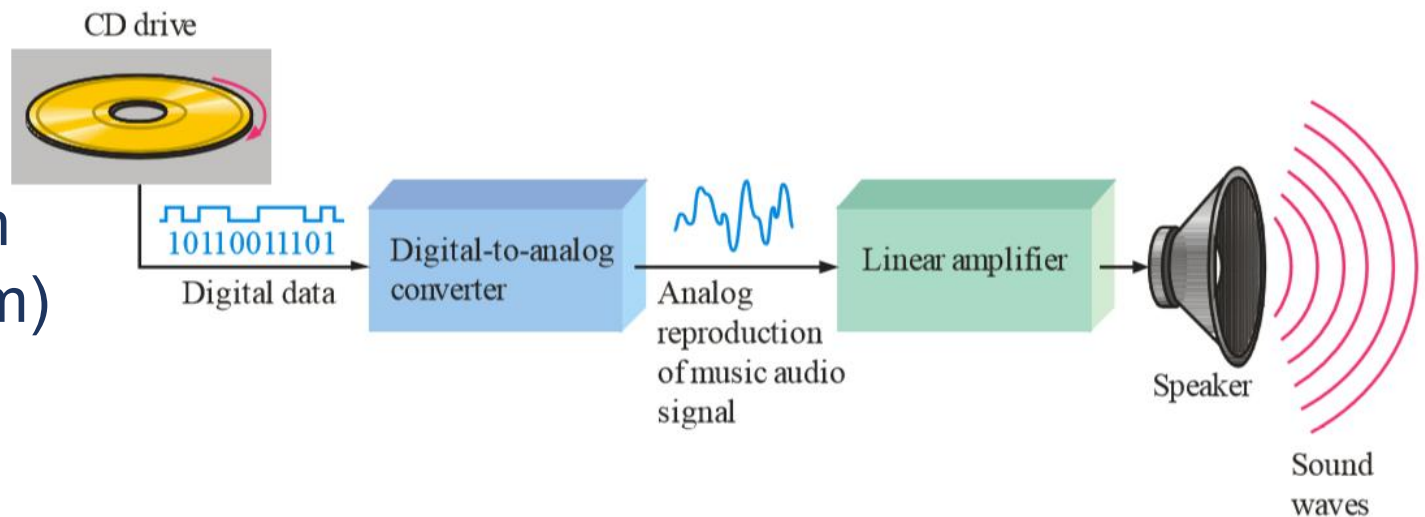


Analog & Digital (Hybrid) Systems: Comparison?

Analog system



Digital system (hybrid system)



How to Represent Digital Signals?

We need to know how to represent bits ...

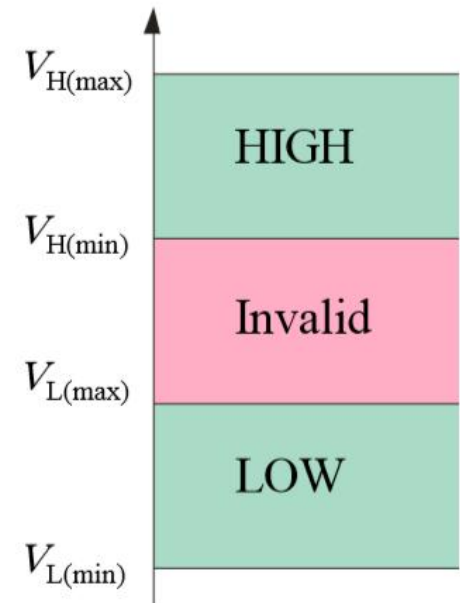
<u>Decimal</u>	<u>Binary</u>	<u>Octal</u>	<u>Hex</u>
0	0000	0	0
1	0001	1	1
2	0010	2	2
3	0011	3	3
4	0100	4	4
5	0101	5	5
6	0110	6	6
7	0111	7	7
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F



Binary Digits and Logic Levels

Digital electronics uses circuits that have two states, which are represented by two different voltage levels called HIGH and LOW. The voltages represent numbers in the binary system.

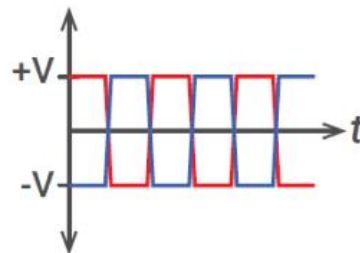
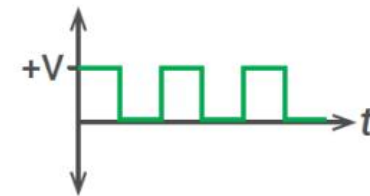
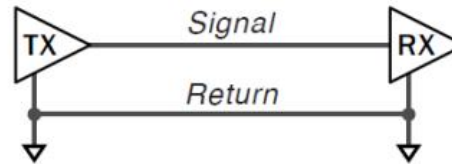
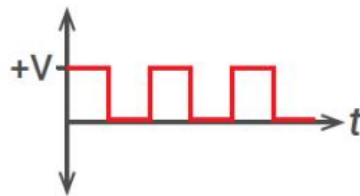
In binary, a single number is called a *bit* (for *binary digit*). A bit can have the value of either a 0 or a 1, depending on if the voltage is HIGH or LOW.



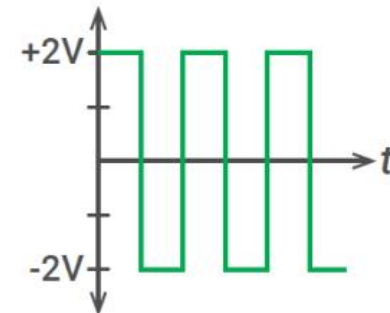
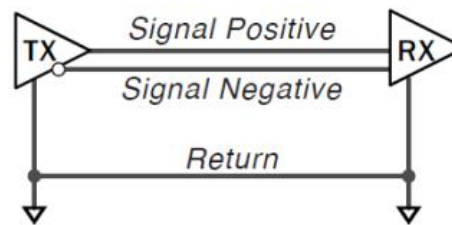
Binary Digits and Logic Levels

Examples of binary logic levels

Technology	L voltage	H voltage	Notes
CMOS ^[3] [4]	0 V to 30% V_{DD}	70% V_{DD} to V_{DD}	V_{DD} = supply voltage
TTL ^[3]	0 V to 0.8 V	2 V to V_{CC}	V_{CC} = 5 V $\pm 5\%$ (7400 commercial family) or $\pm 10\%$ (5400 military family)



— Signal Positive
— Signal Negative



— Output

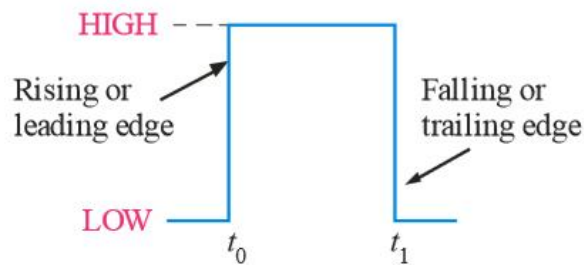
https://en.wikipedia.org/wiki/Logic_level

<https://sparxeng.com/blog/hardware/mastering-differential-signals>

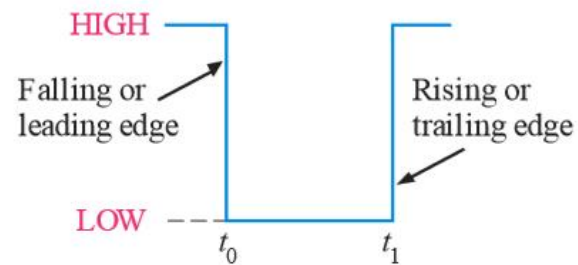


Digital Waveforms

Digital waveforms change between the LOW and HIGH levels. A positive going pulse is one that goes from a normally LOW logic level to a HIGH level and then back again. Digital waveforms are made up of a series of pulses.



(a) Positive-going pulse

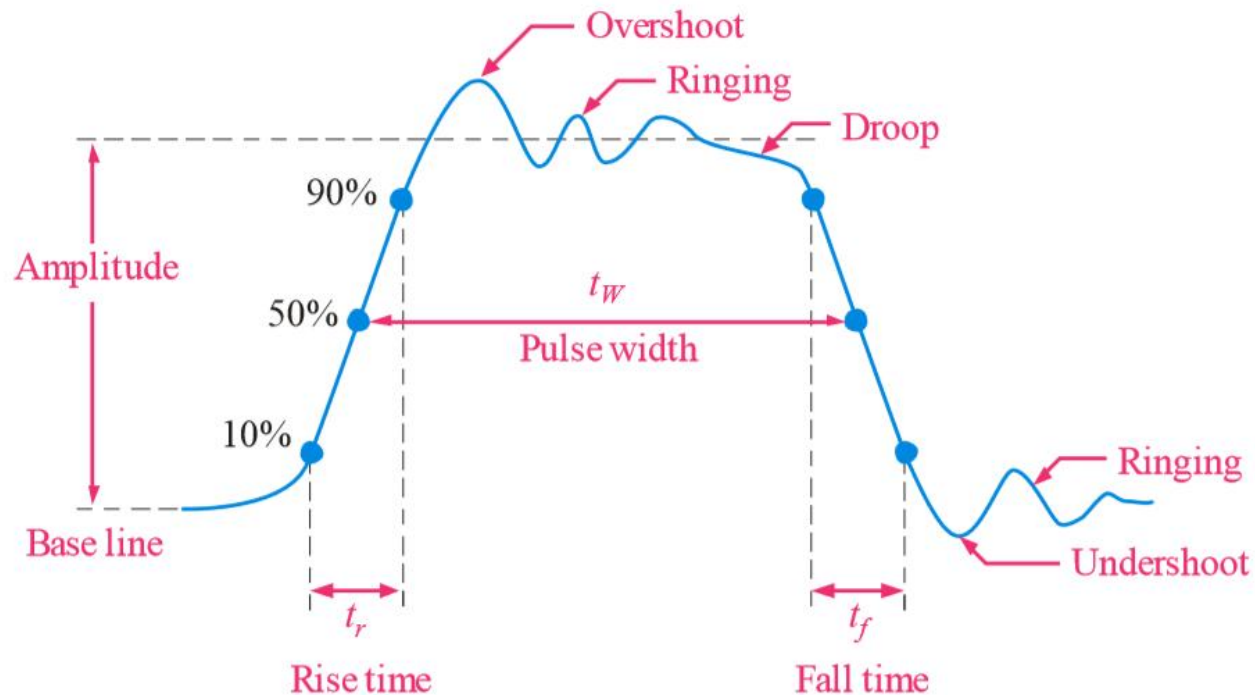


(b) Negative-going pulse



Pulse Definitions

Actual pulses are not ideal but are described by the rise time, fall time, amplitude, and other characteristics.



Periodic Pulse Waveforms

Periodic pulse waveforms are composed of pulses that repeats in a fixed interval called the **period**. The **frequency** is the rate it repeats and is measured in hertz.

$$f = \frac{1}{T} \qquad T = \frac{1}{f}$$

The **clock** is a basic timing signal that is an example of a periodic wave.

Example

What is the period of a repetitive wave if $f = 3.2 \text{ GHz}$?

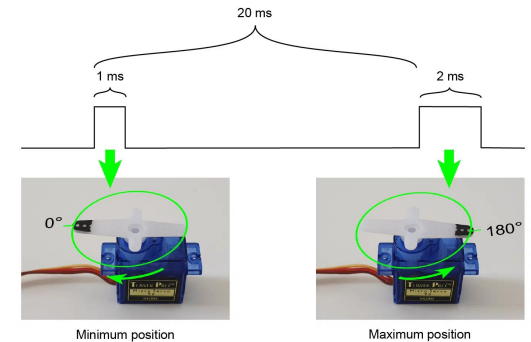
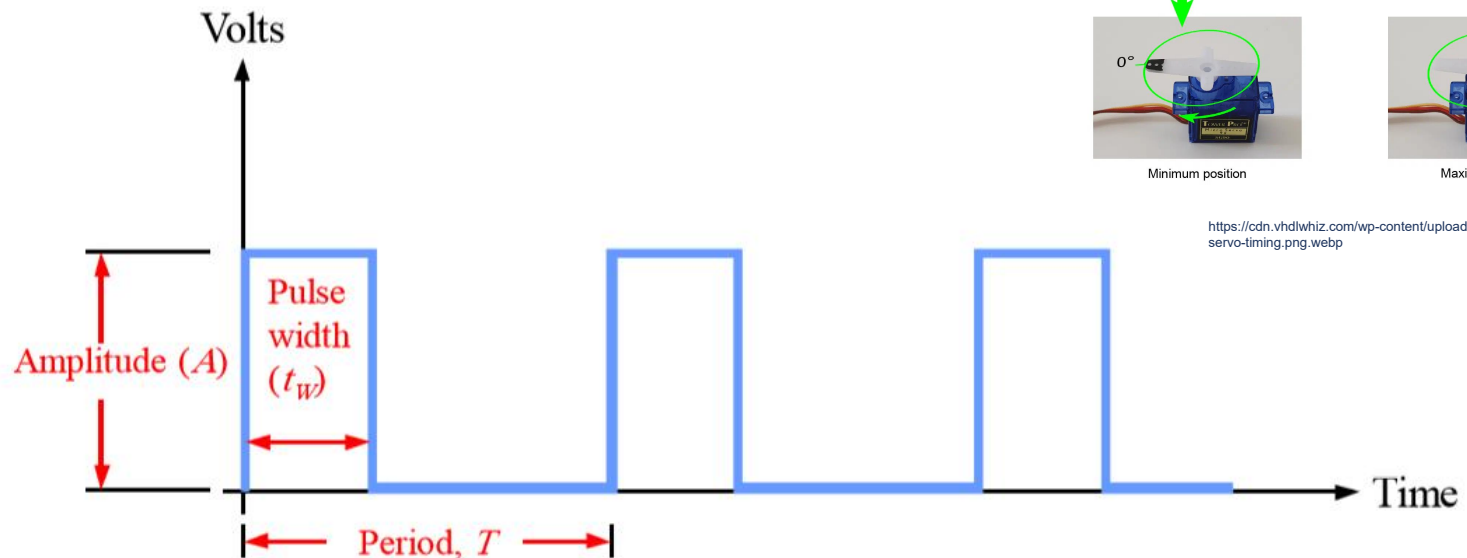
Solution

$$T = \frac{1}{f} = \frac{1}{3.2 \text{ GHz}} = 313 \text{ ps}$$



Pulse Definitions

In addition to frequency and period, repetitive pulse waveforms are described by the amplitude (A), pulse width (t_W) and duty cycle. Duty cycle is the ratio of t_W to T .

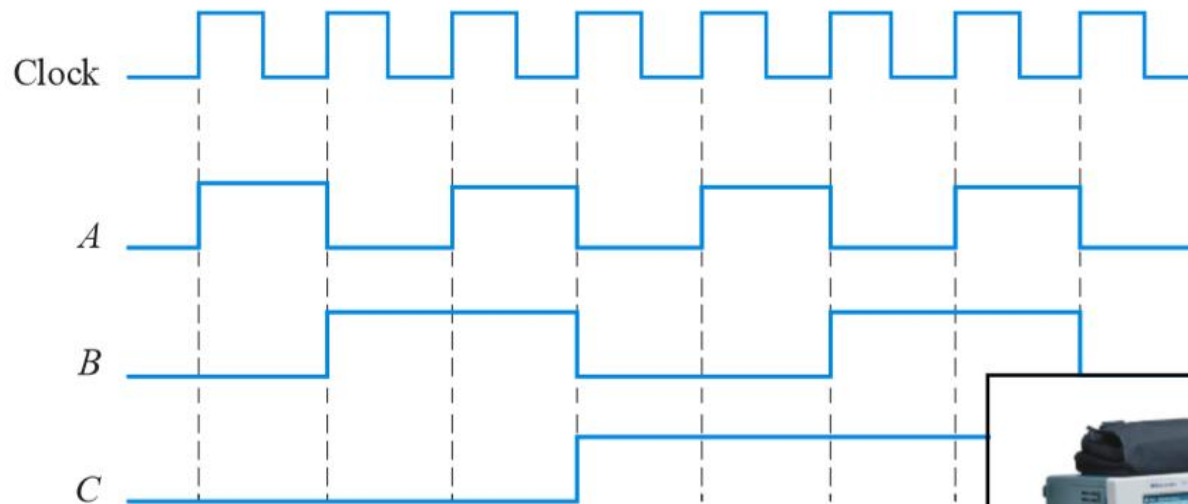


<https://cdn.vhdlwhiz.com/wp-content/uploads/2020/07/rc-servo-timing.png.webp>



Timing Diagrams

A timing diagram is used to show the relationship between two or more digital waveforms,



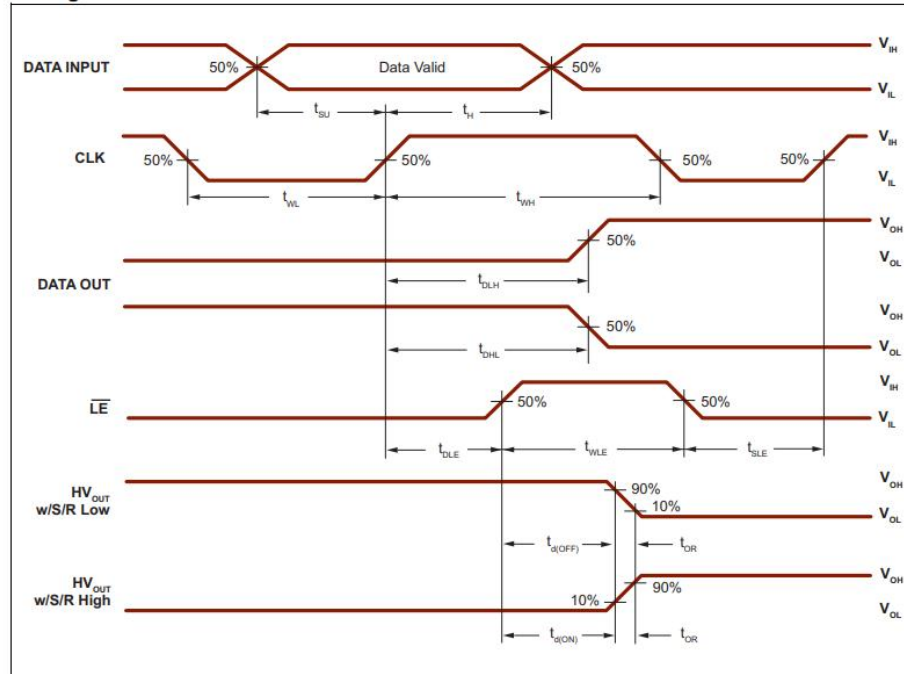
A diagram like this can be observed directly on a logic analyzer.



Example: HV513 Datasheet

In practise, AC characteristics need to be considered

Timing Waveforms



AC ELECTRICAL CHARACTERISTICS

Electrical Specifications: Over typical operating conditions unless otherwise specified, $T_j = 25^\circ\text{C}$.

Parameter	Sym.	Min.	Typ.	Max.	Unit	Conditions
Clock Frequency	f_{CLK}	0	—	8	MHz	
Output Switching Frequency (SOA Limited)	f_{OUT}	—	300	—	Hz	$C_L = 50 \text{ nF}$, $V_{\text{PP}} = 200\text{V}$
Clock Width High and Low	t_{WL} , t_{WH}	62	—	—	ns	
Data Set-Up Time before Clock Rises	t_{SU}	15	—	—	ns	
Data Hold Time after Clock Rises	t_{H}	30	—	—	ns	
Latch Enable Pulse Width	t_{WLE}	80	—	—	ns	
Latch Enable Delay Time after Rising Edge of Clock	t_{DLE}	35	—	—	ns	
Latch Enable Set-Up Time before Clock Rises	t_{SLE}	40	—	—	ns	
HV _{OUT} Rise/Fall Time	t_{OR} , t_{OF}	—	—	1000	μs	$C_L = 100 \text{ nF}$, $V_{\text{PP}} = 200\text{V}$
Delay Time for Output to Start Rise/Fall	$t_{\text{DON/OFF}}$	—	—	500	ns	
Delay Time Clock to Data Low to High	t_{DLH}	—	—	110	ns	$C_L = 15 \text{ pF}$
Delay Time Clock to Data High to Low	t_{DHL}	—	—	110	ns	$C_L = 15 \text{ pF}$
All Logic Inputs	t_r , t_f	—	—	5	ns	
Output Short-Circuit Detection	t_{SD}	—	—	500	ns	$C_L = 15 \text{ pF}$, short to output fall of SHORT
Output Short-Circuit Clear	t_{SC}	—	—	3000	ns	Short clear to output rise of SHORT
Output High-Z State	$t_{\text{HI-Z}}$	—	—	500	ns	

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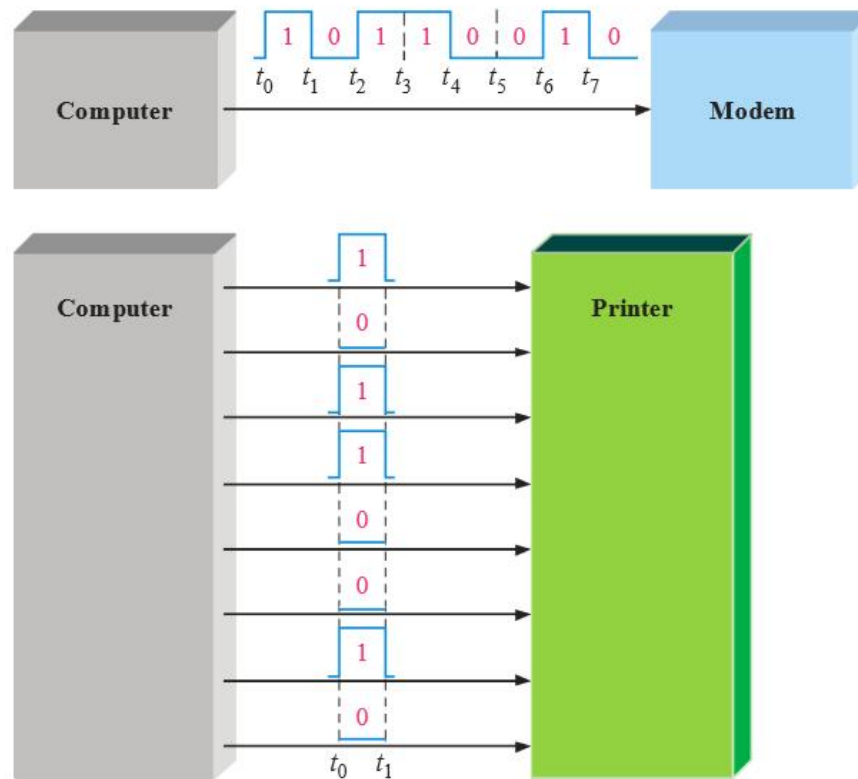
DS20005846B-page 5

<https://ww1.microchip.com/downloads/en/DeviceDoc/HV513-8-Channel-Serial-to-Parallel-Converter-with-High-Voltage-Push-Pull-Outputs-Polarity-Hi-Z-and-Short-Circuit-Detect-Data-Sheet.pdf>



Serial and Parallel Data

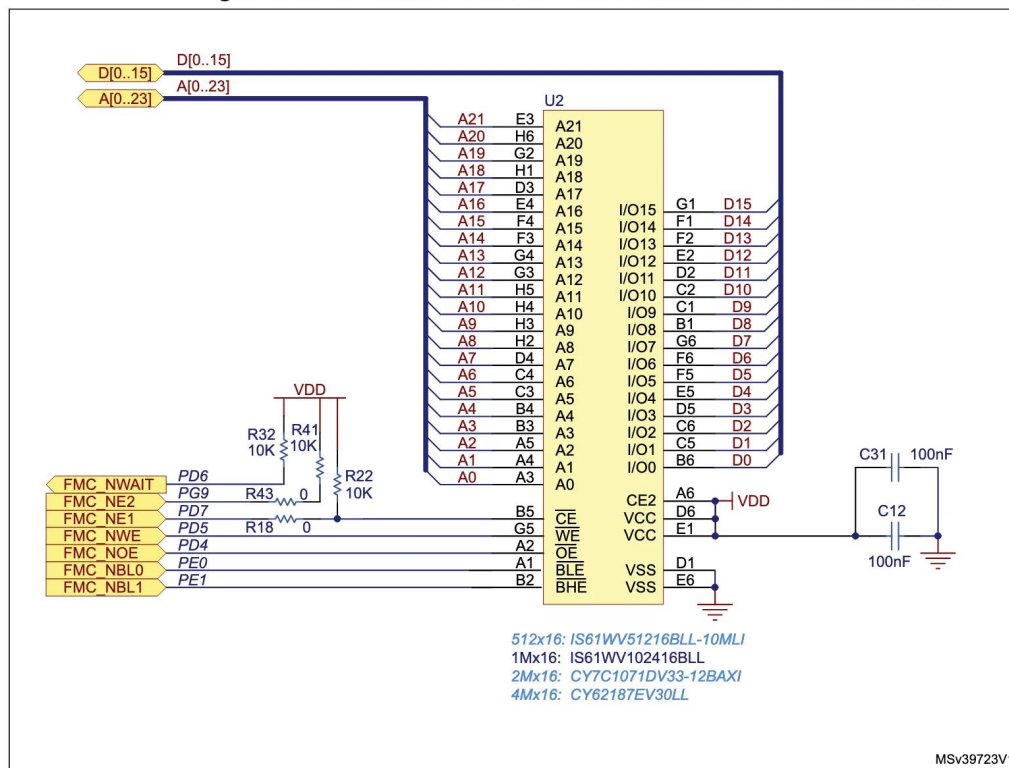
Data can be transmitted by either serial transfer or parallel transfer.



Serial and Parallel Data

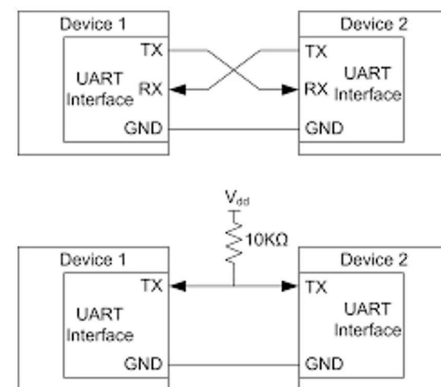
SRAM: parallel data

Figure 8. 16-bit SRAM: IS61WV102416BLL connection to STM32L476/486



https://www.reddit.com/r/AskElectronics/comments/uuya42h/external_sram_access_on_stm32_board_using_fmc/

UART: serial data



<https://microcontrollerslab.com/wp-content/uploads/2021/03/UART-Communication-STM32F4-discovery-board.jpg>

